

Visualization Critique and Redesign Report

Original visualization and context:

Introduction of the Visualization

Source: <https://public.tableau.com/app/profile/matt.chambers/viz/SportsCity/SportsCity>

Matt Chambers's 2018 Tableau visualization *Sports City* asks which North American cities have won the most championships in MLB, NBA, NFL, and NHL. Each city is represented by a flower whose four colored petals represent the leagues. Petal size encodes championship count, while interaction reveals combined totals and team-level details.

Data Source

Main:

https://en.wikipedia.org/wiki/List_of_North_American_cities_by_number_of_major_sports_championships

Additional sources:

https://en.wikipedia.org/wiki/List_of_World_Series_champions

https://en.wikipedia.org/wiki/List_of_NBA_champions

[https://en.wikipedia.org/wiki/List_of_NFL_champions_\(1920–1969\)](https://en.wikipedia.org/wiki/List_of_NFL_champions_(1920–1969))

https://en.wikipedia.org/wiki/List_of_Super_Bowl_champions

https://en.wikipedia.org/wiki/List_of_St Stanley Cup champions

Intended Message & Audience

The visualization communicates that championships are concentrated among relatively few cities. New York leads with 48 championships, followed by Boston with 37. It also shows differences in composition: some cities succeed across multiple leagues, while others are dominated by one sport. Boston, Chicago, Los Angeles, and New York had won championships in all four leagues by 2018.

The primary audience is North American professional sports fans rather than statistical experts, reflected in the playful sports-themed design and interactive exploration.

Tasks

Viewers should be able to compare total championships across cities and within individual leagues, examine league composition, identify balanced or unusually successful cities, highlight a league across cities, and inspect exact totals and team-level details.

Critique:

Strengths

1. Engaging multivariate summary

The four-petal flower provides a memorable and compact visual encoding of four quantitative attributes. From a **data/task abstraction** perspective, each city is an item and its MLB, NFL, NBA, and NHL championship counts are quantitative attributes. The glyph supports an overview task by showing whether success is distributed across multiple sports or concentrated in only one or two.

2. Consistent encoding and interaction

Each league maintains a consistent color and petal position, creating a stable categorical encoding. Selecting a league highlights corresponding petals across cities, while tooltips reveal exact totals and team-level breakdowns. This follows the **overview first, details on demand** principle: overall patterns remain visible without excessive labels, while precise information is available through interaction.

Weaknesses and Opportunities for Improvement

1. Data abstraction hides temporal patterns

Problem: Championships through 2018 are aggregated into cumulative totals, removing championship year from the displayed abstraction. Raw counts also do not account for differences in franchise age or opportunities.

Principle: Data/task abstraction should match the questions users need to answer. Temporal attributes should be retained when temporal comparison is an important task.

Why it matters: Cities with similar totals may have very different histories. One may have accumulated titles decades ago, while another succeeded recently. The visualization therefore cannot reveal whether success is recent, sustained, or historically concentrated.

Improvement: Preserve championship-level city, league, team, and year data. A year-range filter could recompute rankings, while a linked timeline could reveal when championships occurred. A normalized measure such as championships per franchise-season could optionally supplement raw totals.

2. Visual encoding does not prioritize ranking

Problem: Cities are arranged alphabetically rather than by championship count.

Principle: Visual encoding and spatial arrangement should reflect the primary task. For quantitative comparison and ranking, spatial position is one of the most effective channels. **Although alphabetical ordering is expressive for lookup, it is less effective for the primary ranking task because it uses spatial position for nominal organization rather than quantitative comparison.**

Why it matters: Identifying the most successful cities requires scanning many flowers, estimating their sizes, and remembering previous observations.

Improvement: Sort cities by total championships and provide controls to reorder them by total, individual league, or city name.

3. Separate interaction states hinder comparison

Problem: City totals and league highlighting appear in separate interaction states.

Principle: Interaction should support comparison without requiring users to mentally integrate information across states.

Why it matters: Users may want to understand both which cities rank highly and whether that success is distributed across leagues. Switching states increases reliance on working memory.

Improvement: Display total and league composition simultaneously, such as with a total label and four league segments in an aligned row. Highlighting or sorting a league would then preserve the remaining information.

Redesign rationale:

1. Descending stacked bars

The flower grid is replaced with descending horizontal stacked bars. Bar length and position enable more accurate quantitative comparison, while descending order directly supports ranking. Colored segments preserve league composition and numeric labels show exact totals. This addresses both alphabetical ordering and separated interaction states.

2. Temporal filtering

Start- and end-year controls plus meaningful presets allow rankings and league segments to be recomputed for a shared period. This reveals changes in leadership over time and partially addresses unequal historical windows.

3. Linked timeline

Selecting a city displays each championship as a colored dot on a common year axis, separated by league. This reveals clusters, droughts, dynasties, and cross-league success that

cumulative totals hide. Tooltips provide year, league, and team details on demand.

Original vs. redesign:

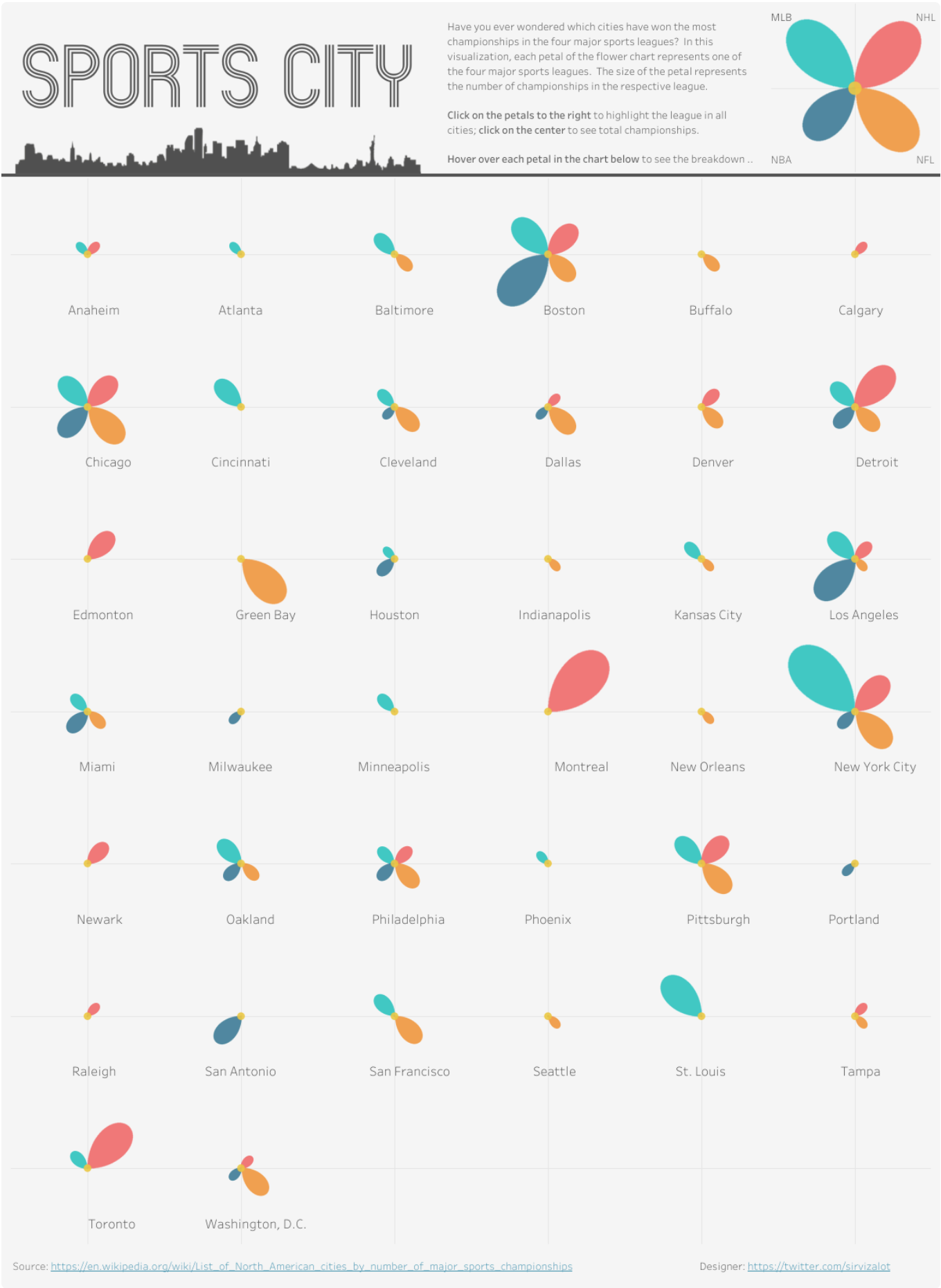
Improvements

The redesign makes rank, exact total, league composition, and timing accessible within a coordinated view. Users can identify leaders without scanning alphabetically, compare leagues along a shared axis, examine common time periods, and move from city-level summaries to championship-level evidence.

Limitations & Tradeoffs

The redesign sacrifices visual novelty: stacked bars are less distinctive than flowers and many cities require vertical scrolling. More importantly, it still primarily reports achievements rather than a fully opportunity-adjusted success rate. Such normalization would require additional data describing which franchises were active in each season.

Figures and references:



Start year

1903

End year

2018

Cities shown

Top 15

Reset view

Quick ranges: All years Four-league era 1980–1999 2000–2018

1903–2018

shared comparison window

384

championship events

New York City · 54

44 cities with a title in this window

MLB NBA NFL NHL

